

1 Summary

Model (Run-name): model1 (test)

Iteration: 23

TAC: 154870.2470796738

Took: 2.5388033390045166s

Total Time: 37.20231008529663s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

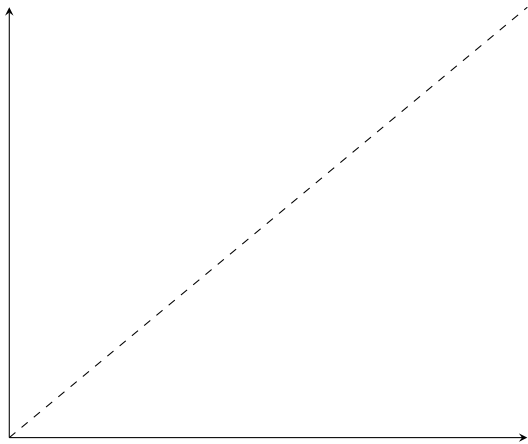
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

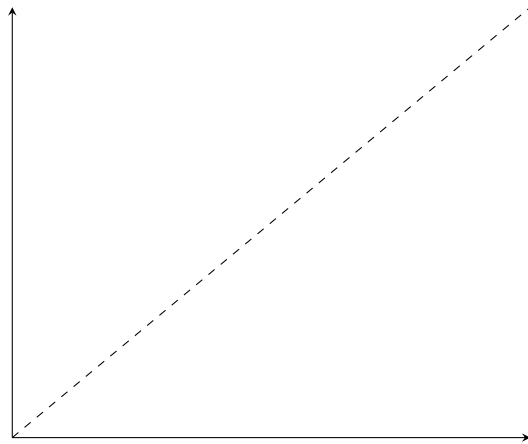
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052069189713074004	$b_{1,1,1}^{H,out}$	905.0460555623276
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017847395759361194	$b_{1,2,2}^{H,out}$	3865.1002822772152
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028680209240537154	$b_{2,1,2}^{H,out}$	9356.176149778303
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.01465404022260934	$b_{1,1,1}^{C,in}$	8572.650423415767
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009566826397397695	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022517062821569487	$b_{2,1,2}^{C,in}$	4143.293955471636
$A_{1,1,1}^\beta$	71.18527282143235	$q_{1,1,1}$	684.8997177228433	$b_{1,1,1}^{C,out}$	9257.55014113861
$A_{1,2,2}^\beta$	69.60484346150977	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	138.86359281280824	$q_{2,1,2}$	2422.6504234158283	$b_{2,1,2}^{C,out}$	6565.944378887465
$A_{cu,1}^\beta$	4.813467676447643	$q_{cu,1}$	165.100282277215		
$A_{cu,2}^\beta$	37.821651654621526	$q_{cu,2}$	1977.3495765841812		
$A_{hu,1}^\beta$	13.294574825457861	$q_{hu,1}$	492.44985886139057		
$A_{1,1,1}$	71.18527282143235	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.60484346150977	$t_{1,2}^{(H)}$	581.5100282277216		
$A_{2,1,2}$	138.86359281280824	$t_{1,3}^{(H)}$	386.5100282277215		
$A_{cu,1}$	4.813467676447643	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.821651654621526	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.294574825457861	$t_{2,3}^{(H)}$	468.86747882920906		
$dt_{1,1,1}$	32.82999059075951	$t_{1,1}^{(C)}$	617.1700094092406		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.5100282277159		
$dt_{1,2,2}$	81.51002822772182	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.51002822772189	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.48997177228432	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.86747882921052	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.5100282277215	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.86747882920906	$t_{1,2,2}^H$	386.5100282277219		
$dt_{hu,1}$	62.82999059075937	$t_{2,1,2}^H$	468.65031894324017		
$f_{1,1,1}^H$	2.4460704204387245	$t_{1,1,1}^C$	617.1700094092407		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	19.964112835922258	$t_{2,1,2}^C$	649.8205080141946		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1589.9457732851708		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5815.100282277215		
$f_{2,1,2}^C$	10.105595013345454	$b_{2,1,2}^{H,in}$	11778.826573194126		

3 *RecLMTD* Tangent Points

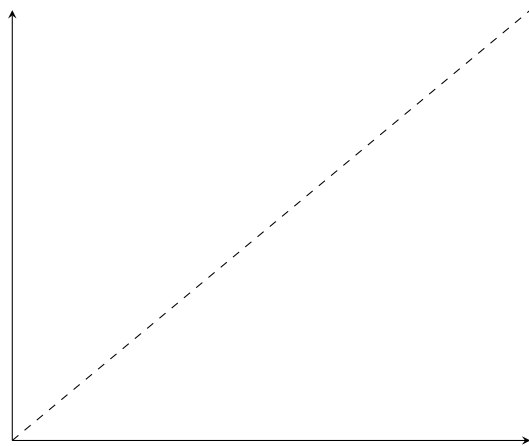
- $(1, 1, 1)$
 - Value: 0.052069
 - Error: 0.000000798979590
 - Relative error: 0.000015344339603
- $(1, 2, 2)$
 - Value: 0.017847
 - Error: 0.000000139565051
 - Relative error: 0.000007819850123
- $(2, 1, 2)$
 - Value: 0.028680
 - Error: 0.000000617331282
 - Relative error: 0.000021524180285
- $cu, 1$
 - Value: 0.014654
 - Error: 0.000000091293915
 - Relative error: 0.000006229909609
- $cu, 2$
 - Value: 0.009567
 - Error: 0.000000631260848
 - Relative error: 0.000065979999108
- $hu, 1$
 - Value: 0.022517
 - Error: 0.000000005313187
 - Relative error: 0.000000235962665



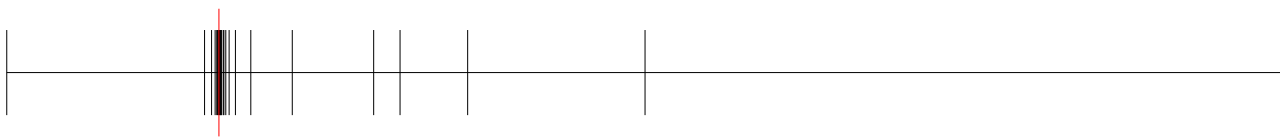
(a) *RecLMTD* - (1, 1, 1)



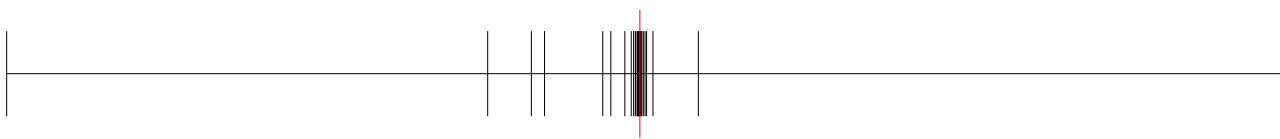
(b) *RecLMTD* - (1, 2, 2)



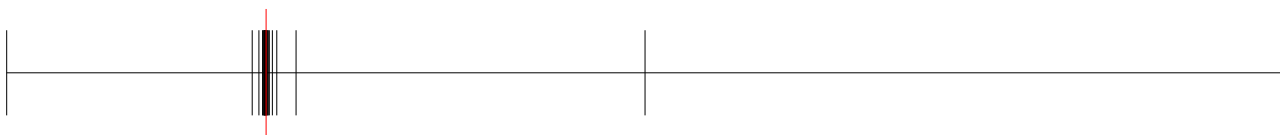
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1589.9457732851708
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5815.100282277215
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.5640223858990211e-16
- (2, 1, 2)
 - Value: 11778.826573194126
 - Absolute error: 5.4569682106375694e-12
 - Relative error: -4.632862345606106e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 905.0460555623276
 - Absolute error: 4.547473508864641e-13
 - Relative error: -5.024576905137916e-16
- (1, 2, 2)
 - Value: 3865.1002822772152
 - Absolute error: 3.637978807091713e-12
 - Relative error: -9.412378829530157e-16
- (2, 1, 2)
 - Value: 9356.176149778303
 - Absolute error: 0.011698195497956476
 - Relative error: -1.250316441700117e-06

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8572.650423415767
 - Absolute error: 2.9103830456733704e-11
 - Relative error: 3.394962936694367e-15
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4143.293955471636
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9257.55014113861
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6565.944378887465
 - Absolute error: 0.878506470389766
 - Relative error: -0.00013377952865891745



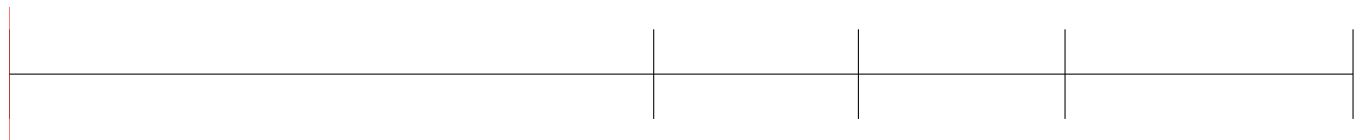
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



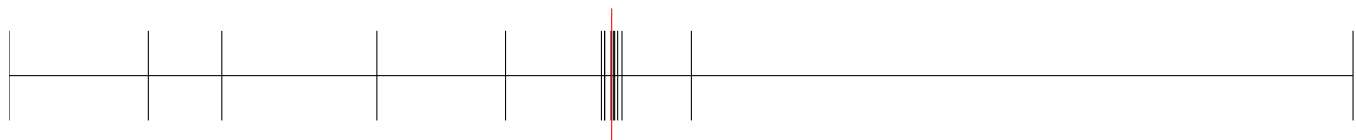
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$

5 Area Breakpoints

5.1 A

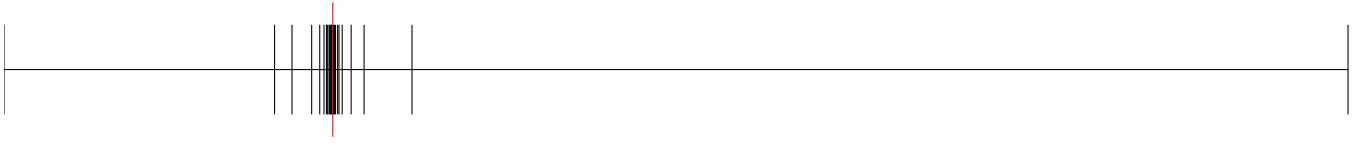
- $(1, 1, 1)$
 - Value: 71.185273
 - Error: 0.139073851650778
 - Relative error: 0.001949879082499
- $(1, 2, 2)$
 - Value: 69.604843
 - Error: 0.000000000001123
 - Relative error: -0.000000000000016
- $(2, 1, 2)$
 - Value: 138.863593
 - Error: 0.100649307675553
 - Relative error: 0.000724282060908

5.2 A_{CU}

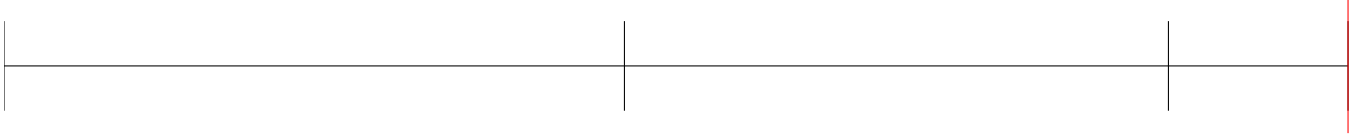
- $cu, 1$
 - Value: 4.813468
 - Error: 0.025304678061286
 - Relative error: 0.005229565726048
- $cu, 2$
 - Value: 37.821652
 - Error: 0.012268597675877
 - Relative error: 0.000324275084212

5.3 A_{HU}

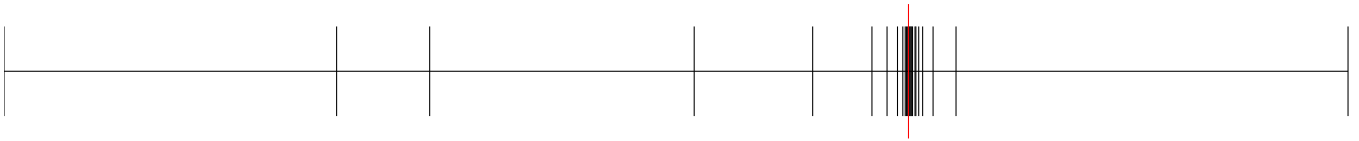
- $hu, 1$
 - Value: 13.294575
 - Error: 0.011654464688089
 - Relative error: 0.000875865313453



(a) $A^\beta - (1, 1, 1)$



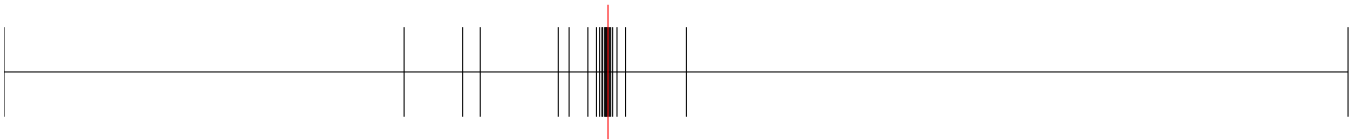
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

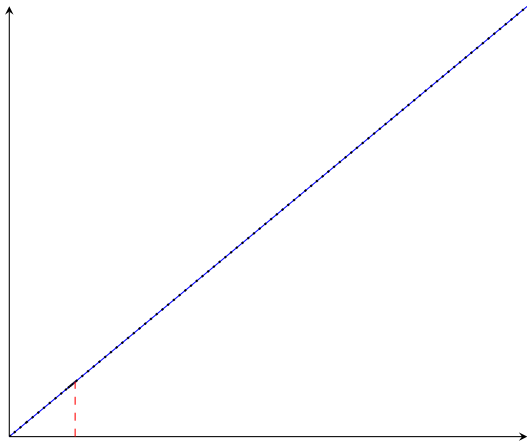


(a) $A^\beta - hu, 1$

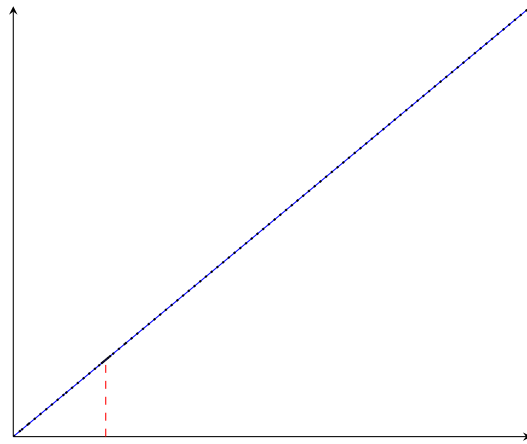
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.185273
 - $\hat{q}^\beta$: 71.185273
 - q^β : 71.185273
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.604843
 - $\hat{q}^\beta$: 69.604843
 - q^β : 69.604843
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 68.10969007738701, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 70.08419437796795, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 138.863593
 - $\hat{q}^\beta$: 138.863593
 - q^β : 138.863593
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 137.02923097052596, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 144.67005099128556, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.813468
 - $\hat{q}^\beta$: 4.813468
 - q^β : 4.813468
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.707108999269609, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 5.081618324442152, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000

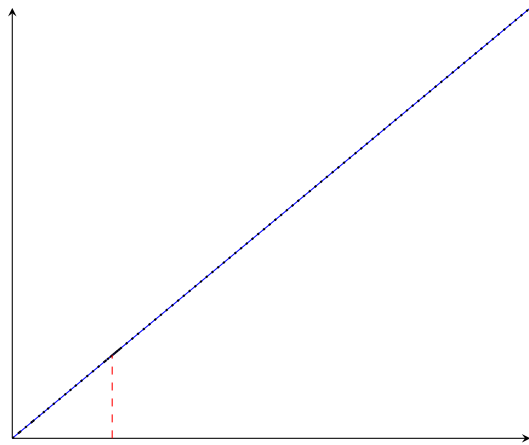
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.821652
 - $\hat{q}^\beta$: 37.821652
 - q^β : 37.821652
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, 37.612770977560544, 37.6551299822246, 37.682669717644174, 37.71812813120871, 37.76995088018704, **37.78573239590169**, **37.88148462263016**, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.294575
 - $\hat{q}^\beta$: 13.294575
 - q^β : 13.294575
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.266498143734903, **13.269097009439808**, **13.30871858348944**, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



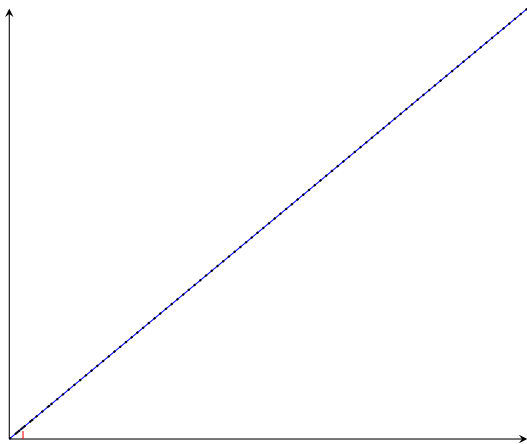
(a) $q^\beta - (1, 1, 1)$



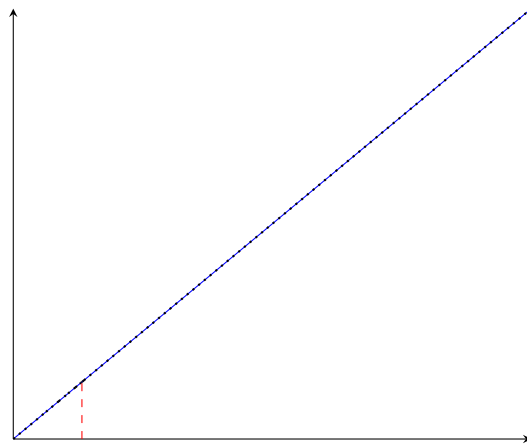
(b) $q^\beta - (1, 2, 2)$



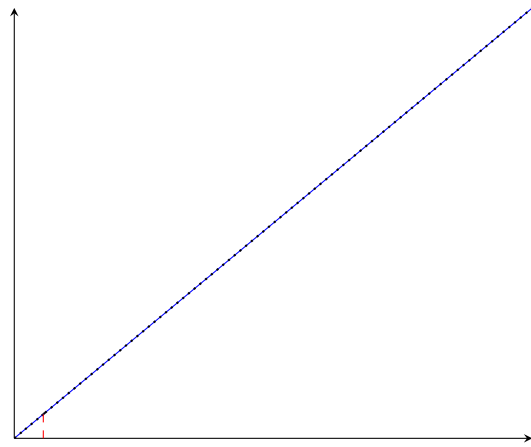
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	21.173426805867166
$z_{1,2,1}$	0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	2006.706044528364
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	0.0	$RecLMTD_{1,2,1}$	0.01607047381606835	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	0.0	$RecLMTD_{2,2,1}$	0.02011172883803818	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	2006.7060445283623
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	436.91512735787		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	576.9059677174985		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	410.0		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	31.691916800738852	$t_{2,2,2}^C$	350.0		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4910.054226714829		
$f_{1,2,1}^H$	7.5539295795612755	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	21.173426805867166		
$f_{2,2,2}^H$	0.035887164077742284	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.894404986654546	$b_{1,2,1}^{H,out}$	4910.0542267148885		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		